

DATABASE MANAGEMENT SYSTEM LAB

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Subject: Database Management System

LAB 4

Question: Implement SQL Functions: Conversion Functions, and Date Functions.

Output:

```
1 CREATE TABLE employees (
2   emp_id INT PRIMARY KEY,
3   emp_name VARCHAR(50),
4   department VARCHAR(30),
5   salary DECIMAL(10,2),
6   bonus DECIMAL(10,2),
7   hire_date DATE,
8   email VARCHAR(60)
9 );
10
11 INSERT INTO employees VALUES (101, 'Aditi Rao', 'Analytics', 55000.00, 5000.00, '2023-06-15', 'aditi@corp.co');
12 INSERT INTO employees VALUES (102, 'Ravi Kumar', 'Finance', 62000.50, NULL, '2022-01-10', 'ravi@corp.co');
13 INSERT INTO employees VALUES (103, 'Sneha Iyer', 'Analytics', 48000.75, 3000.00, '2024-03-01', 'sneha@corp.co');
14 INSERT INTO employees VALUES (104, 'Arjun Mehta', 'HR', 41000.00, NULL, '2021-11-20', NULL);
15 INSERT INTO employees VALUES (105, 'Priya Nair', 'Finance', 70000.00, 8000.00, '2020-08-05', 'priya@corp.co');
16
17 CREATE TABLE contractors (
18   con_id INT PRIMARY KEY,
19   con_name VARCHAR(50),
20   department VARCHAR(30),
21   fee DECIMAL(10,2)
22 );
23
24 INSERT INTO contractors VALUES (201, 'Neha Singh', 'Analytics', 45000.00);
25 INSERT INTO contractors VALUES (202, 'Ravi Kumar', 'Finance', 62000.50);
26 INSERT INTO contractors VALUES (203, 'Omar Khan', 'IT', 52000.00);
27
28
29 SELECT CAST('1500' AS SIGNED) + 500 AS total;
30 SELECT CAST('2024-12-25' AS DATE) AS my_date;
31 SELECT emp_name, CAST(salary AS CHAR) AS salary_text FROM employees;
32 SELECT STR_TO_DATE('25-12-2024', '%d-%m-%Y') AS my_date;
```

Output:

total
2000

emp_name	salary_text
Aditi Rao	55000.00
Ravi Kumar	62000.50
Sneha Iyer	48000.75
Arjun Mehta	41000.00
Priya Nair	70000.00

my_date
2024-12-25

```
12 INSERT INTO employees VALUES (102, 'Ravi Kumar', 'Finance', 62000.50, NULL, '2022-01-10', 'ravi@corp.co');
13 INSERT INTO employees VALUES (103, 'Sneha Iyer', 'Analytics', 48000.75, 3000.00, '2024-03-01', 'sneha@corp.co');
14 INSERT INTO employees VALUES (104, 'Arjun Mehta', 'HR', 41000.00, NULL, '2021-11-20', NULL);
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18   con_id INT PRIMARY KEY,
19   con_name VARCHAR(50),
20   department VARCHAR(30),
21   fee DECIMAL(10,2)
22 );
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24 INSERT INTO contractors VALUES (201, 'Neha Singh', 'Analytics', 45000.00);
25 INSERT INTO contractors VALUES (202, 'Ravi Kumar', 'Finance', 62000.50);
26 INSERT INTO contractors VALUES (203, 'Omar Khan', 'IT', 52000.00);
27
28
29 SELECT CAST('1500' AS SIGNED) + 500 AS total;
30 SELECT CAST('2024-12-25' AS DATE) AS my_date;
31 SELECT emp_name, CAST(salary AS CHAR) AS salary_text FROM employees;
32 SELECT STR_TO_DATE('25-12-2024', '%d-%m-%Y') AS my_date;
33 SELECT emp_name, DATE_FORMAT(hire_date, '%d-%m-%Y') AS hire_text FROM employees;
34 SELECT emp_name, IFNULL(bonus, 0) AS bonus FROM employees;
35
36
37 SELECT NOW();
38 SELECT CURDATE();
39 SELECT emp_name, hire_date, DATE_ADD(hire_date, INTERVAL 6 MONTH) AS after_6_months FROM employees;
40 SELECT emp_name, DATEDIFF(CURDATE(), hire_date) AS days_worked FROM employees;
41 SELECT emp_name, YEAR(hire_date) AS yr, MONTH(hire_date) AS mon, DAY(hire_date) AS dy FROM employees;
42 SELECT emp_name, DAYNAME(hire_date) AS day_name FROM employees;
43 SELECT emp_name, LAST_DAY(hire_date) AS month_end FROM employees;
```

Output:

emp_name	hire_text
Aditi Rao	15-06-2023
Ravi Kumar	10-01-2022
Sneha Iyer	01-03-2024
Arjun Mehta	20-11-2021
Priya Nair	05-08-2020

emp_name	bonus
Aditi Rao	5000.00
Ravi Kumar	0.00
Sneha Iyer	3000.00
Arjun Mehta	0.00
Priya Nair	8000.00

NOW()
2026-09-29 17:41:58

The screenshot shows the OneCompiler MySQL interface. The left pane contains SQL queries, and the right pane shows the output of the executed queries.

```

1 CREATE TABLE employees (
2   emp_name VARCHAR(50),
3   department VARCHAR(30),
4   salary DECIMAL(10,2),
5   bonus DECIMAL(10,2),
6   hire_date DATE,
7   email VARCHAR(50)
8 );
9
10
11 INSERT INTO employees VALUES (101, 'Aditi Rao', 'Analytics', 55000.00, 5000.00, '2023-06-15', 'aditirao');
12 INSERT INTO employees VALUES (102, 'Ravi Kumar', 'Finance', 62000.50, NULL, '2022-01-10', 'ravikumar');
13 INSERT INTO employees VALUES (103, 'Sneha Iyer', 'Analytics', 48000.75, 3000.00, '2024-03-01', 'snehai');
14 INSERT INTO employees VALUES (104, 'Arjun Mehta', 'HR', 41000.00, NULL, '2021-11-20', NULL);
15 INSERT INTO employees VALUES (105, 'Priya Nair', 'Finance', 70000.00, 8000.00, '2020-08-05', 'priyanair');
16
17 CREATE TABLE contractors (
18   con_id INT PRIMARY KEY,
19   con_name VARCHAR(50),
20   department VARCHAR(30),
21   fee DECIMAL(10,2)
22 );
23
24 INSERT INTO contractors VALUES (201, 'Neha Singh', 'Analytics', 45000.00);
25 INSERT INTO contractors VALUES (202, 'Ravi Kumar', 'Finance', 62000.50);
26 INSERT INTO contractors VALUES (203, 'Omar Khan', 'IT', 52000.00);
27
28
29 SELECT CAST('1500' AS SIGNED) + 500 AS total;
30 SELECT CAST('2024-12-25' AS DATE) AS my_date;
31 SELECT emp_name, CAST(salary AS CHAR) AS salary_text FROM employees;
32 SELECT STR_TO_DATE('25-12-2024', '%d-%m-%Y') AS my_date;
33 SELECT emp_name, DATE_FORMAT(hire_date, '%d-%m-%Y') AS hire_text FROM employees;
34 SELECT emp_name, IFNULL(bonus, 0) AS bonus FROM employees;
35
36 SELECT NOW();
37 SELECT CURDATE();
38 SELECT emp_name, hire_date, DATE_ADD(hire_date, INTERVAL 6 MONTH) AS after_6_months FROM employees;
39 SELECT emp_name, DATEDIFF(CURDATE(), hire_date) AS days_worked FROM employees;
40 SELECT emp_name, YEAR(hire_date) AS yr, MONTH(hire_date) AS mon, DAY(hire_date) AS dy FROM employees;
41 SELECT emp_name, DAYNAME(hire_date) AS day_name FROM employees;
42 SELECT emp_name, LAST_DAY(hire_date) AS month_end FROM employees;

```

The output shows the results of these queries, including employee and contractor data, date calculations, and conditional logic.

LAB 5

Question: Implement SQL Operators: Demonstrate the use of different types of operators in SQL with examples, including Arithmetic Operators, Logical Operators, Comparison Operators.

Output:

The screenshot shows the OneCompiler MySQL interface with more SQL queries and their outputs.

```

23
24 INSERT INTO contractors VALUES (201, 'Neha Singh', 'Analytics', 45000.00);
25 INSERT INTO contractors VALUES (202, 'Ravi Kumar', 'Finance', 62000.50);
26 INSERT INTO contractors VALUES (203, 'Omar Khan', 'IT', 52000.00);
27
28
29 SELECT CAST('1500' AS SIGNED) + 500 AS total;
30 SELECT CAST('2024-12-25' AS DATE) AS my_date;
31 SELECT emp_name, CAST(salary AS CHAR) AS salary_text FROM employees;
32 SELECT STR_TO_DATE('25-12-2024', '%d-%m-%Y') AS my_date;
33 SELECT emp_name, DATE_FORMAT(hire_date, '%d-%m-%Y') AS hire_text FROM employees;
34 SELECT emp_name, IFNULL(bonus, 0) AS bonus FROM employees;
35
36 SELECT NOW();
37 SELECT CURDATE();
38 SELECT emp_name, hire_date, DATE_ADD(hire_date, INTERVAL 6 MONTH) AS after_6_months FROM employees;
39 SELECT emp_name, DATEDIFF(CURDATE(), hire_date) AS days_worked FROM employees;
40 SELECT emp_name, YEAR(hire_date) AS yr, MONTH(hire_date) AS mon, DAY(hire_date) AS dy FROM employees;
41 SELECT emp_name, DAYNAME(hire_date) AS day_name FROM employees;
42 SELECT emp_name, LAST_DAY(hire_date) AS month_end FROM employees;
43
44
45
46
47 SELECT emp_name, salary, salary + 5000 AS addition FROM employees;
48 SELECT emp_name, salary, salary - 2000 AS subtraction FROM employees;
49 SELECT emp_name, salary, salary * 12 AS yearly_salary FROM employees;
50 SELECT emp_name, salary, salary / 2 AS division FROM employees;
51 SELECT emp_name, salary, MOD(salary, 7) AS remainder FROM employees;
52
53
54 SELECT * FROM employees WHERE salary = 55000;
55 SELECT * FROM employees WHERE department <> 'Finance';
56 SELECT * FROM employees WHERE salary > 55000;
57 SELECT * FROM employees WHERE salary < 50000;
58 SELECT * FROM employees WHERE salary >= 55000;
59 SELECT * FROM employees WHERE salary <= 48000.75;
60
61
62 SELECT * FROM employees WHERE department = 'Finance' AND salary > 65000;
63 SELECT * FROM employees WHERE department = 'HR' OR department = 'Finance';
64 SELECT * FROM employees WHERE NOT department = 'Analytics';

```

The output shows the results of these queries, including contractor data, arithmetic operations, and conditional logic.

OneCompiler

MySQLRun

AI Agent

86 ms

queries.sql

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Ready

STORM

Input for the program (Optional)

Output

emp_name	salary	division
Aditi Rao	55000.00	27500.000000
Ravi Kumar	62000.50	31000.250000
Sneha Iyer	48000.75	24000.375000
Arjun Mehta	41000.00	20500.000000
Priya Nair	70000.00	35000.000000

emp_name	salary	remainder
Aditi Rao	55000.00	1.00
Ravi Kumar	62000.50	1.50
Sneha Iyer	48000.75	1.75
Arjun Mehta	41000.00	1.00
Priya Nair	70000.00	0.00

emp_id	emp_name	department	salary	bonus	hire_date	email
101	Aditi Rao	Analytics	55000.00	5000.00	2023-06-15	aditi@corp.com

emp_id	emp_name	department	salary	bonus	hire_date	email
101	Aditi Rao	Analytics	55000.00	5000.00	2023-06-15	aditi@corp.com
103	Sneha Iyer	Analytics	48000.75	3000.00	2024-01-01	sneha@corp.com
104	Arjun Mehta	HR	41000.00	NUL	2021-11-20	NUL

emp_id	emp_name	department	salary	bonus	hire_date	email
102	Ravi Kumar	Finance	62000.50	NUL	2022-01-10	ravi@corp.com
105	Priya Nair	Finance	70000.00	8000.00	2020-08-05	priya@corp.com

OneCompiler

MySQLRun

AI Agent

86 ms

queries.sql

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Ready

STORM

Input for the program (Optional)

Output

emp_id	emp_name	department	salary	bonus	hire_date	email
104	Arjun Mehta	HR	41000.00	NUL	2021-11-20	NUL

emp_id	emp_name	department	salary	bonus	hire_date	email
101	Aditi Rao	Analytics	55000.00	5000.00	2023-06-15	aditi@corp.com
102	Ravi Kumar	Finance	62000.50	NUL	2022-01-10	ravi@corp.com
105	Priya Nair	Finance	70000.00	8000.00	2020-08-05	priya@corp.com

emp_id	emp_name	department	salary	bonus	hire_date	email
103	Sneha Iyer	Analytics	48000.75	3000.00	2024-01-01	sneha@corp.com
104	Arjun Mehta	HR	41000.00	NUL	2021-11-20	NUL

emp_id	emp_name	department	salary	bonus	hire_date	email
105	Priya Nair	Finance	70000.00	8000.00	2020-08-05	priya@corp.com

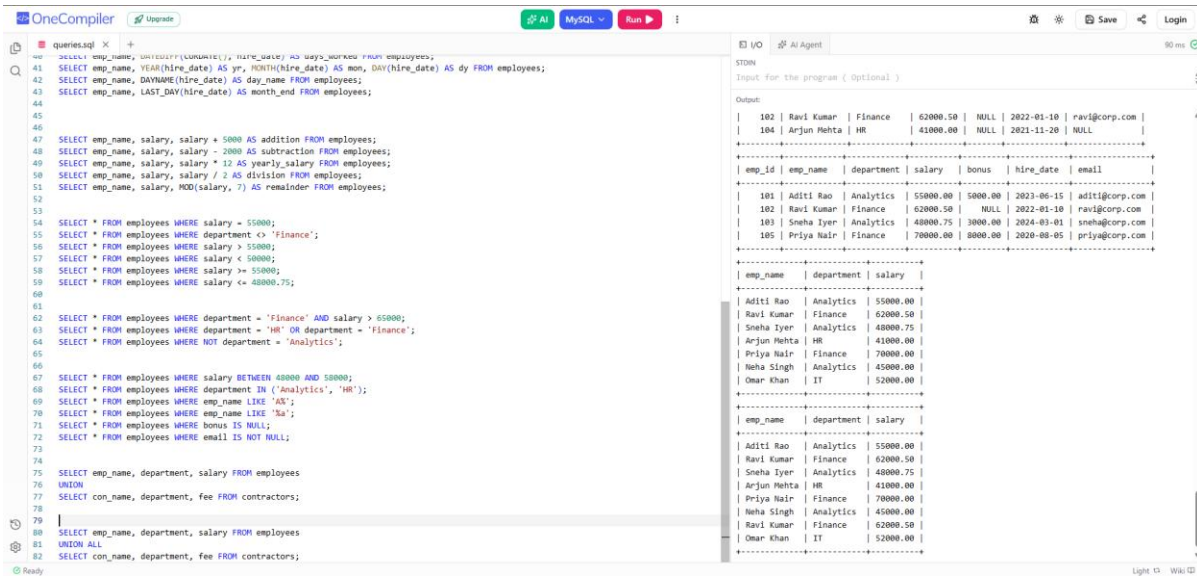
emp_id	emp_name	department	salary	bonus	hire_date	email
102	Ravi Kumar	Finance	62000.50	NUL	2022-01-10	ravi@corp.com
104	Arjun Mehta	HR	41000.00	NUL	2021-11-20	NUL
105	Priya Nair	Finance	70000.00	8000.00	2020-08-05	priya@corp.com

emp_id	emp_name	department	salary	bonus	hire_date	email
102	Ravi Kumar	Finance	62000.50	NUL	2022-01-10	ravi@corp.com
104	Arjun Mehta	HR	41000.00	NUL	2021-11-20	NUL
105	Priya Nair	Finance	70000.00	8000.00	2020-08-05	priya@corp.com

LAB 6

Question: Implement SQL Operators: Demonstrate the use of different types of operators in SQL with examples, Special Operators, and Set Operations (UNION).

Output:



The screenshot shows the OneCompiler MySQL interface. The left pane contains a list of SQL queries, and the right pane shows the output of the selected query (Query 41).

Queries:

```
40 SELECT emp_name, DOWDAY(hire_date) AS days_worked FROM employees;
41 SELECT emp_name, YEAR(hire_date) AS yr, MONTH(hire_date) AS mon, DAY(hire_date) AS dy FROM employees;
42 SELECT emp_name, DAYNAME(hire_date) AS day_name FROM employees;
43 SELECT emp_name, LAST_DAY(hire_date) AS month_end FROM employees;
44
45
46 SELECT emp_name, salary, salary + 5000 AS addition FROM employees;
47 SELECT emp_name, salary, salary - 2000 AS subtraction FROM employees;
48 SELECT emp_name, salary, salary * 12 AS yearly_salary FROM employees;
49 SELECT emp_name, salary, salary / 2 AS division FROM employees;
50 SELECT emp_name, salary, MOD(salary, 7) AS remainder FROM employees;
51
52
53
54 SELECT * FROM employees WHERE salary = 55000;
55 SELECT * FROM employees WHERE department <> 'Finance';
56 SELECT * FROM employees WHERE salary > 55000;
57 SELECT * FROM employees WHERE salary < 55000;
58 SELECT * FROM employees WHERE salary >= 55000;
59 SELECT * FROM employees WHERE salary <= 48000.75;
60
61
62 SELECT * FROM employees WHERE department = 'Finance' AND salary > 65000;
63 SELECT * FROM employees WHERE department = 'HR' OR department = 'Finance';
64 SELECT * FROM employees WHERE NOT department = 'Analytics';
65
66
67 SELECT * FROM employees WHERE salary BETWEEN 48000 AND 58000;
68 SELECT * FROM employees WHERE department IN ('Analytics', 'HR');
69 SELECT * FROM employees WHERE emp_name LIKE 'A%';
70 SELECT * FROM employees WHERE emp_name LIKE '%a';
71 SELECT * FROM employees WHERE bonus IS NULL;
72 SELECT * FROM employees WHERE email IS NOT NULL;
73
74
75 SELECT emp_name, department, salary FROM employees
76 UNION
77 SELECT con_name, department, fee FROM contractors;
78
79
80 SELECT emp_name, department, salary FROM employees
81 UNION ALL
82 SELECT con_name, department, fee FROM contractors;
```

Output (Query 41):

emp_id	emp_name	department	salary	bonus	hire_date	email
102	Ravi Kumar	Finance	62000.50	NULL	2022-01-10	ravi@corp.com
104	Arjun Mehta	HR	41000.00	NULL	2021-11-20	NULL
101	Aditi Rao	Analytics	55000.00	5000.00	2023-06-15	aditi@corp.com
102	Ravi Kumar	Finance	62000.50	NULL	2022-01-10	ravi@corp.com
103	Sneha Iyer	Analytics	48000.75	3000.00	2024-03-01	sneha@corp.com
105	Priya Nair	Finance	70000.00	8000.00	2020-08-05	priya@corp.com
Aditi Rao	Analytics	55000.00				
Ravi Kumar	Finance	62000.50				
Sneha Iyer	Analytics	48000.75				
Arjun Mehta	HR	41000.00				
Priya Nair	Finance	70000.00				
Neha Singh	Analytics	45000.00				
Omar Khan	IT	52000.00				
Aditi Rao	Analytics	55000.00				
Ravi Kumar	Finance	62000.50				
Sneha Iyer	Analytics	48000.75				
Arjun Mehta	HR	41000.00				
Priya Nair	Finance	70000.00				
Neha Singh	Analytics	45000.00				
Ravi Kumar	Finance	62000.50				
Omar Khan	IT	52000.00				